

Zinc-Doped High-Nickel, Low-Cobalt Layered Oxide Cathodes for High-Energy-Density Lithium-Ion Batteries

Zehao Cui, Qiang Xie, and Arumugam Manthiram*

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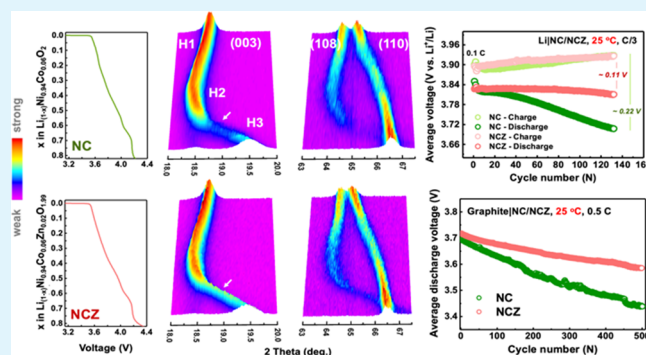
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ABSTRACT: High-Ni layered oxides with Ni contents greater than 90% are promising cathode candidates for high-energy-density Li-ion batteries. However, drastic electrode–electrolyte reactions and mechanical degradation issues limit their cycle life and practical viability. We demonstrate here that $\text{Li}_{0.94}\text{Ni}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}\text{O}_{1.99}$ (NCZ), obtained by incorporating 2 mol % Zn^{2+} into an ultrahigh-Ni baseline cathode material $\text{Li}_{0.94}\text{Ni}_{0.94}\text{Co}_{0.06}\text{O}_2$ (NC), delivers superior cell performance. NCZ retains 74% of the initial capacity after 500 cycles in a full cell assembled with a graphite anode, outperforming NC (62% retention). NCZ also possesses a higher average discharge voltage relative to NC with an outstanding average voltage retention of over 99% after 130 cycles in half cells. Bulk structural investigations unveil that Zn doping promotes a smoother phase transition, suppresses anisotropic lattice distortion, and maintains the mechanical integrity of cathode particles. Furthermore, NCZ shows an enhanced interphase stability after long-term cycling, in contrast to the seriously degraded surface chemistry in NC. This work provides a practically viable approach for designing higher-energy-density high-Ni layered oxide cathodes for lithium-ion batteries.

KEYWORDS: lithium-ion batteries, high-nickel layered oxides, zinc doping, phase transitions, electrode–electrolyte interphases



1. INTRODUCTION

The groundbreaking discovery of layered oxide cathodes in the 1980s opened the door for Li-ion batteries, which have catalyzed the advancement of electrical power sources.¹ In recent years, the burgeoning global electric vehicle (EV) market calls for advanced cathode materials with a high energy density at affordable prices.^{2,3} Despite great success in portable devices, LiCoO_2 is not suitable for EV applications due to the high cost, toxicity, and scarcity of Co.⁴ Li-rich Mn-based cathode materials, $\text{Li}_2\text{MnO}_3 \cdot \text{LiNi}_x\text{Co}_y\text{Mn}_{1-x-y}\text{O}_2$, are able to deliver an appealing capacity of over 250 mA h g^{-1} with a lower cost.^{5,6} Nevertheless, the large irreversible capacity loss during the first cycle and the rapid voltage decay of Li-rich cathodes hinder their commercialization.^{7,8} In contrast, Ni-based layered oxide cathodes, $\text{LiNi}_x\text{Co}_y\text{Mn}_z\text{O}_2$ ($\text{M} = \text{Mn}$ or Al ; NMC or NCA), are receiving increasing attention due to their high volumetric and gravimetric energy densities and increased calendar life.⁹ The development of Ni-based layered oxide cathodes is moving gradually from low-Ni, high-Co compositions to high-Ni, low-Co or Co-free compositions to enable higher-energy and lower-cost cathodes. Over the past decade, the development of NMC cathodes proceeded along the path of NMC-333 $\rightarrow$ NMC-532 $\rightarrow$ NMC-622 $\rightarrow$ NMC-811, where the three numbers refer to the atomic ratio of Ni, Mn, and Co in the cathode, respectively. In both the industry and academia,

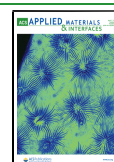
increasing work is now focused on ultrahigh-Ni oxide cathodes ($\text{Ni} > 0.90$) due to their attractive capacity of over 220 mA h g^{-1} , outstanding rate performance, and significantly reduced cost.¹⁰ However, the deterioration of NMC cathodes is accelerated with an increasing Ni content.¹¹

To begin with, the lattice distortion of Ni-based NMC cathodes becomes more significant for high-Ni cathodes compared to low-Ni cathodes as more Li^+ is extracted from the lattice during charge, resulting in a mechanical failure of high-Ni NMC cathodes after long-term operation.^{11–13} An increased amount of highly reactive Ni^{4+} in high-Ni cathodes promotes parasitic reactions with the electrolyte at high voltages and raises safety issues.¹⁴ Meanwhile, surface lattice reconstruction of high-Ni cathodes from a layered structure to a rock-salt NiO phase during cycling results in impedance buildup and material fatigue.^{15,16} Therefore, for the viable applications of ultrahigh-Ni cathodes, it is of great importance

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to ameliorate these performance degradations without compromising capacity.

Lattice doping with ions, such as Al, Mg, Mn, Zr, Ti, Ga, and so on, provides an effective tuning of the high-Ni cathode chemistry.^{17–22} The incorporation of these doping ions greatly enhances the cycling and thermal stability of high-Ni cathodes. However, the performance enhancement brought on by these dopants is commonly accompanied by a sharp decrease in capacity. For example, introducing 5 mol % of Al, Mn, and Mg into LiNiO_2 causes a capacity drop of, respectively, approximately 30, 35, and 50 mA h g^{-1} .⁴ Additionally, the synthesis of Ti-, Al-, Zr-, and Ga-doped high-Ni cathodes is typically realized through a wet chemistry approach followed by subsequent calcination, which could result in compositional inhomogeneity, poor batch reproducibility, accumulation of metal oxides impurities, and irregular cell degradation behaviors.^{3,23} In contrast, synthesis through co-precipitation demonstrates great advantage in promoting the homogeneity of the doping elements in high-Ni cathodes, as revealed by recent synthesis of boron-doped $\text{LiNi}_{0.9-x}\text{Co}_{0.1}\text{B}_x\text{O}_2$ and Fe-doped $\text{LiNi}_{0.85}\text{Fe}_{0.052}\text{Al}_{0.91}\text{O}_2$.^{24,25} Among the diverse dopant candidates, Zn is proposed as another promising element due to its low cost and similar K_{sp} value of $\text{Zn}(\text{OH})_2$ (3.0×10^{-17}) to that of $\text{Ni}(\text{OH})_2$ (5.5×10^{-15}), which makes it feasible to synthesize a Zn-doped high-Ni cathode via hydroxide co-precipitation.²⁶ More importantly, the ionic sizes of Li^+ and Zn^{2+} are very close ($r_{\text{Li}^+} = 76$ pm and $r_{\text{Zn}^{2+}} = 74$ pm), which suggests that Zn ions are likely to reside in Li sites.²⁷ Thus, Zn^{2+} could act as a pillaring ion to ameliorate the lattice collapse, similar to Mg^{2+} , which has also been shown to reside in the Li sites.²⁸ To our knowledge, however, a comprehensive study of the impact of Zn^{2+} on the chemistry and performance of ultrahigh-Ni cathodes remains unexplored.

Here, we present a Zn-doped ultrahigh-Ni cathode, $\text{LiNi}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}\text{O}_{1.99}$ (NCZ). Through co-precipitation, homogeneous doping of Zn in NCZ is achieved without triggering impurity accumulation at the particle surface. NCZ is able to deliver a high discharge capacity of 218 mA h g^{-1} at a C/3 rate compared to 223 mA h g^{-1} for the baseline material, $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (NC). The cyclability and voltage retention of NCZ outperform those of NC in both half cells and pouch-type full cells with a graphite anode. Differential scanning calorimetry (DSC) demonstrates a superior thermal stability of NCZ in comparison to NC. A combination of ex situ and in situ X-ray diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM), and X-ray photoelectron spectroscopy (XPS) characterizations shows mitigated electrolyte–electrode reactions on both the cathode and anode as well as suppressed particle cracking and better structural integrity in NCZ. Therefore, homogeneous Zn-doping realized by co-precipitation is demonstrated to benefit the development of high-energy-density ultrahigh-Ni cathodes with an extended service life.

2. EXPERIMENTAL SECTION

2.1. Materials' Preparation. Both the $\text{Ni}_{0.94}\text{Co}_{0.06}(\text{OH})_2$ and $\text{Ni}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}(\text{OH})_2$ precursors were synthesized by the co-precipitation method. For $\text{Ni}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}(\text{OH})_2$, a 2.0 M aqueous solution of $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$, $\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$, and $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (molar ratio of Ni/Co/Zn = 94:4:2) was first prepared. Then, this transition-metal solution, along with a 4.0 M KOH solution and a 14.8 M NH_4OH solution, was pumped separately into a 10 L continuously stirred tank reactor for the hydroxide co-precipitation reaction. During co-precipitation, the pH (~ 11.6), temperature (~ 50 °C), and

stirring rate (600 rpm) were carefully controlled. Afterward, the material was washed, filtered, and dried to remove all impurities and residual moisture. By adjusting the metal solution composition to Ni/Co = 94:6 and following the same procedures as above, $\text{Ni}_{0.94}\text{Co}_{0.06}(\text{OH})_2$ was also synthesized. $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ and $\text{LiNi}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}\text{O}_{1.99}$ cathode materials were then prepared by firing the as-prepared $\text{Ni}_{0.94}\text{Co}_{0.06}(\text{OH})_2$ and $\text{Ni}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}(\text{OH})_2$ precursors with $\text{LiOH} \cdot \text{H}_2\text{O}$ (molar ratio of the precursor/ $\text{LiOH} \cdot \text{H}_2\text{O}$ = 1:1.03) at 660 and 640 °C, respectively, under a flowing oxygen atmosphere for 15 h.

2.2. Electrochemical Tests. To prepare the cathode slurries, the active materials (NC/NCZ), conductive black (Super P), and poly(vinylidene) fluoride (PVDF) binder (90:5:5 by weight) were mixed with a proper amount of *N*-methyl-2-pyrrolidone (NMP). Afterward, the slurries were cast onto an Al foil with an active-material loading of 8–9 mg cm^{-2} . After drying, the electrodes were punched into round disks (1.27 cm diameter) for coin cell tests and rectangular sheets (4.78 cm $\times$ 8.15 cm) for full cell preparation. Coin cells were prepared by stacking the obtained cathode electrodes, Celgard 2500 separator, and fresh Li-metal chips together with the flooded electrolyte in an argon-filled glove box. Similarly, pouch-type full cells were fabricated with the cathode electrodes and graphite electrodes (MTI Corp.) wetted by the electrolyte (1.08 N/P ratio). The electrolyte solution consisted of 1 M LiPF_6 in ethylene carbonate (EC) and ethyl methyl carbonate (EMC) (3:7 by weight) with 2 wt % vinylene carbonate (VC) as a functional additive.

Coin cells were tested on Arbin battery cyclers and cycled between 2.8 and 4.4 V at a C/3 rate after three formation cycles at a C/10 rate ($1\text{C} = 180 \text{ mA g}^{-1}$). Pouch full cells were cycled between 2.5 and 4.3 V at a C/2 rate after three C/10 activation cycles with a constant current–constant voltage (CCCV) mode for all charging steps. Electrochemical impedance spectroscopy (EIS) measurements on pouch full cells (charged to 3.7 V before the EIS tests) were conducted on an impedance spectrometer (Solartron 1260A) by applying a 5 mV potential perturbation and recording the responses from 1 MHz to 1 mHz.

2.3. Characterizations. To study the structural features of the cathode materials, powder X-ray diffraction (XRD) was conducted on the fresh and cycled cathode materials from 10 to 80° in a Rigaku Miniflex 600 (Cu $K\alpha$ radiation). The morphology of the lithiated cathode powders and cycled electrodes was explored by scanning electron microscopy (SEM) (FEI Quanta 650). Chemical compositions of pristine NC and NCZ cathode materials were measured with inductively coupled plasma optical emission spectroscopy (ICP-OES) (Varian 715 ES).

XPS analysis was conducted on an Axis Ultra DLD spectrometer (Kratos) with an Al $K\alpha$ radiation source (1486.6 eV). The cycled anode and cathode electrodes were rinsed in a dimethyl carbonate (DMC) solvent three times to wash away any residual electrolyte component on the surfaces. A JEOL NeoARM transmission electron microscope (TEM) was used to conduct energy dispersive X-ray spectroscopy (EDS) on the fresh cathode samples prepared by a Scios 2 HiVac dual-beam FIB/SEM system.

For the in situ XRD experiments, diffraction patterns were collected from 10 to 80°. By casting slurries onto a Be current collector, a specially designed X-ray transparent cell was used to perform diffraction experiments during the cell operation. The in situ cells were paired with a Li-metal anode and cycled to an identical state of charge (80% of lithium removed).

Thermal stabilities of NC and NCZ were evaluated with differential scanning calorimetry (DSC) measurements on a Netzsch STA 449 system. To prepare the samples, NC and NCZ cathode electrodes were first charged to 220 mA h g^{-1} after three formation cycles in coin half cells at a C/10 rate. Then, the charged cells were carefully disassembled inside a glove box, and the obtained cathode electrodes were washed with dimethyl carbonate (DMC) three times. After the cleaning, the dried cathode electrode materials (active materials along with carbon black and PVDF) were collected and sealed (0.5 N m^{-1}) in a stainless-steel crucible with about 40 wt % electrolyte injected.

The thermal scanning was recorded at a heating rate of $1\text{ }^{\circ}\text{C min}^{-1}$ from 30 to $350\text{ }^{\circ}\text{C}$ under an Ar atmosphere.

3. RESULTS AND DISCUSSION

3.1. Morphology, Elemental Distribution, and Structural Characterization. In Figure 1 and Figure S1, SEM

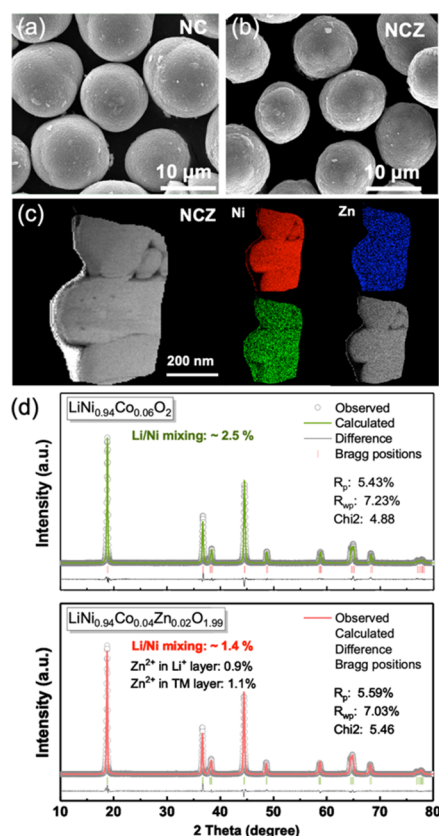


Figure 1. SEM images of calcined (a) NC and (b) NCZ samples. (c) TEM image and EDS maps of primary particles in NCZ. (d) Rietveld refinement results of the XRD patterns of pristine NC and NCZ samples.

images show that both NC and NCZ samples possess a similar spherical morphology with a diameter of $\sim 13\text{ }\mu\text{m}$ on average. EDS tests (TEM in Figure 1c and SEM in Figure S2) illustrate the uniform distribution of Ni, Co, and Zn elements in both primary and secondary particles of the NCZ sample. In addition, ICP-OES (Table S1) analysis reveals that 1.96 mol % of Zn has been successfully incorporated into NCZ and there is a similar Ni content in both samples, agreeing with the designed compositional ratio. XRD patterns of the freshly prepared NC and NCZ materials are shown in Figure S3. A single-phase ($R3m$ space group), layered lattice structure is revealed for both samples, and no other impurity peaks are observed. In summary, structural and compositional results indicate the successful synthesis of NCZ with a homogeneous Zn distribution.

Beyond phase identification, XRD patterns also show that the intensity ratio of the (003) peak to the (104) peak in NCZ (1.28) is similar to that in NC (1.29). Such a similarity implies that it is less likely for Zn^{2+} ions to reside exclusively in the Li^+ layers; otherwise, Zn^{2+} with a large atomic number (Z) would reflect a higher intensity of the (104) peak. As confirmed by Rietveld refinement results summarized in Figure 1d and Table

S2, Zn^{2+} is more likely to occupy both the Li^+ and TM layers, in great agreement with the similar $I(003)/I(104)$ values between NC and NCZ XRD patterns. Former studies suggest that the site preference of foreign ions could play a critical role in regulating the solid-state chemistry of layered oxide cathodes. For instance, Mg^{2+} prefers to occupy the Li^+ site, acting as a pillar to suppress the lattice distortion at high delithiation states.²⁸ Meanwhile, other dopants like Al^{3+} and Ti^{4+} prefer to enter TM sites and perturb local chemical environments.^{18,29} Zn^{2+} , on the other hand, likely resides in both Li^+ and TM sites and thereby could potentially play multiple roles in tuning the structure and altering electrochemical features.

3.2. Electrochemical Performance. The electrochemical performance of NC and NCZ cathodes is compared in Figure 2. The influence of Zn^{2+} on the capacity–discharge profiles is shown in Figure 2a. With the incorporation of 2 mol % of Zn^{2+} into the bulk structure, NCZ delivers a discharge capacity of 225 mA h g^{-1} , slightly lower than that of NC (232 mA h g^{-1}). This is in agreement with the XRD results, indicating that not all the Zn^{2+} ions occupy the Li^+ sites as dopants with a site preference for Li^+ sites that could cause a capacity sacrifice even at low concentrations.⁴ The dQ/dV curves in Figure 2b clearly illustrate that Zn^{2+} plays a critical role in suppressing phase transitions, particularly the H2–H3 phase transition at high voltages. Additionally, the H2–H3 phase transition of NCZ takes place at 17 mV higher than that of NC, which is beneficial for the energy density of NCZ. Furthermore, as shown in Figure 2c, NCZ does not show any compromise on the rate capability until a high rate of 5C.

Figure 2d compares the cycling performance of NC and NCZ in coin half cells. Similar to the formation cycles, NCZ only experiences a minor initial capacity penalty ($\sim 5\text{ mA h g}^{-1}$) even at a C/3 rate compared to NC. However, the cyclability of NCZ greatly outperforms that of NC. The capacity retention of NCZ after 130 cycles is 86.7%, much higher than that of NC (74.7%), which is also accompanied by an increased coulombic efficiency (Figure S4). The evolution of charge and discharge curves (normalized curves in Figure 2e–f and original curves in Figure S5) of NC and NCZ shows significant differences. Initially, the voltage plateaus, which represent the phase transition of H1–H2–H3, are noticeable for both samples; however, the plateaus in NC are indiscernible after 130 cycles, while the plateaus in NCZ remain distinguished. In addition, NCZ delivers a better voltage retention than NC. After 130 cycles, the average discharge voltage retention of NCZ is 99.4%, greatly outperforming that of NC (97.1%), as shown in Figure 2g. Despite a similar average charge/discharge voltage hysteresis at the beginning for both samples, the hysteresis is increased to 0.22 V for NC after 130 cycles, twice that of NCZ. These results suggest that NC may suffer from an increased impedance buildup and irreversible structural changes over NCZ after cycling.

Long-term cycling performance of both samples was examined in pouch-type full cells with a graphite anode. As Figure 2h–i displays, NCZ delivers a discharge capacity of 200 mA h g^{-1} at a C/2 rate, near that of NC (204 mA h g^{-1}). The minor difference in the initial capacities of both samples in pouch full cells is consistent with the results in coin half cells, confirming the negligible influence of Zn doping on the capacity of the high-Ni cathode. After 500 cycles, NCZ delivers a capacity retention of 74%, superior to the 62% of NC.

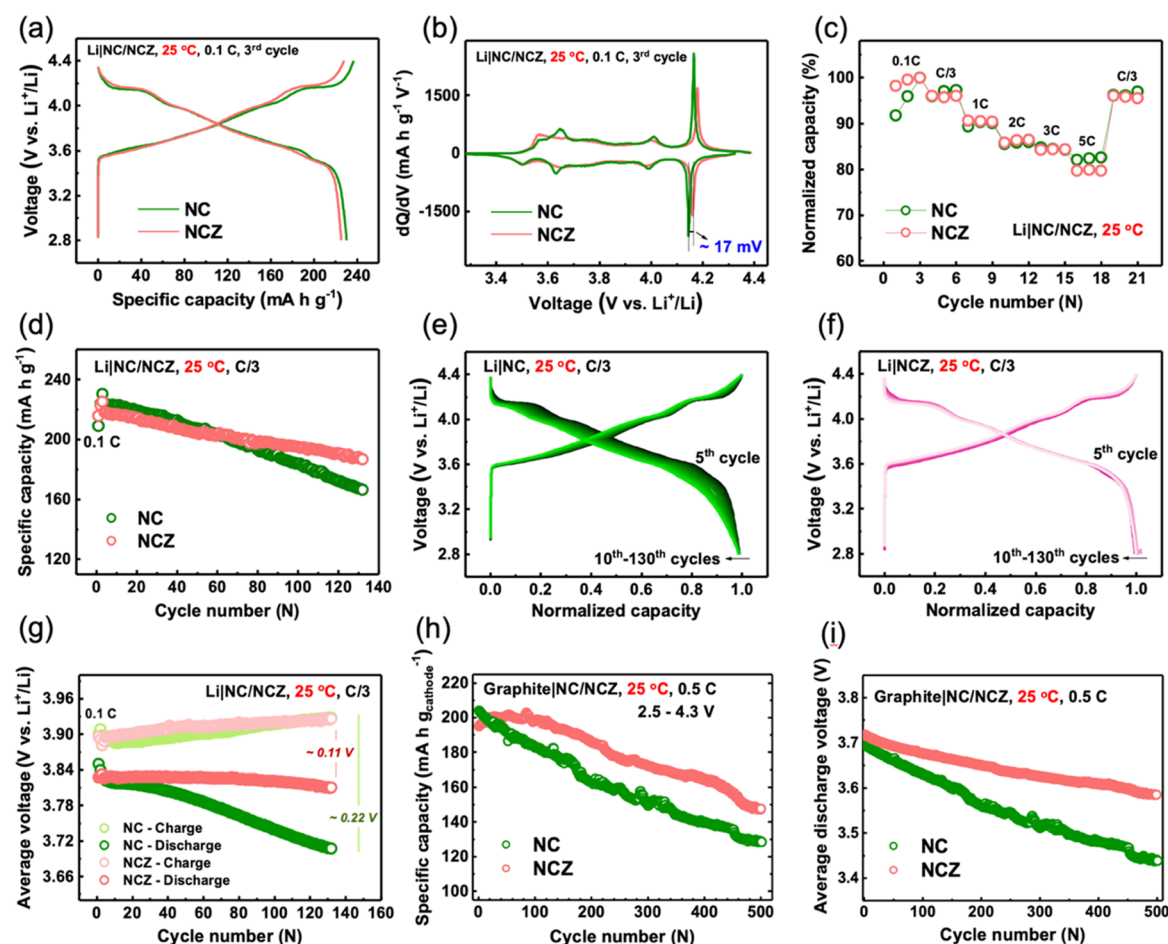


Figure 2. Electrochemical performance comparison of NC and NCZ. (a) Charge–discharge curves, (b) dQ/dV curves, (c) normalized rate performance, (d) cycling performance, (e, f) evolution of the normalized charge–discharge curves over 130 cycles, and (g) evolution of average charge and discharge voltages in half cells (vs Li^+/Li). (h) Cycling performance and (i) average discharge voltage fading of NC and NCZ in pouch full cells cycled between 2.5 and 4.3 V over 500 cycles.

Meanwhile, the average discharge voltage of NCZ fades slightly from 3.71 to 3.60 V, in sharp contrast to the significant voltage drop from 3.70 to 3.45 V for NC. Collectively, electrochemical tests indicate that the incorporation of Zn into high-Ni cathodes improves the energy retention of the high-Ni cathode without inducing a major capacity penalty.

3.3. Phase Transitions and Structural Degradation.

Lattice anisotropic distortions have been recognized as a universal feature for NMC cathodes, and the extent of distortion primarily depends on the degree of delithiation.^{12,30} For ultrahigh-Ni cathodes, more Li^+ ions are extracted at a given voltage compared to low-Ni cathodes. As a result, high-Ni cathodes generally experience increased anisotropic distortions with an increasing Ni content. Such serious distortions lead to a mechanical failure of cathode particles (particle cracking), exacerbation of electrode/electrolyte side reactions by providing more fresh surfaces, rapid buildup of resistance, and ultimately failure of the cathode.³¹

To reveal the differences in the lattice structural stability, both samples were examined by in situ XRD charged to the same state of delithiation. Figure 3b,c, respectively, shows the evolution of the (003) and (108)/(110) Bragg reflection peaks of NC and NCZ upon charging to an identical delithiation state of 80 mol % (80% of Li^+ removed) along with the corresponding charging curves in Figure 3a. For the (003)

peak, both NC and NCZ show a phase transition from H1 to H2 and eventually to H3, consistent with previous studies.^{28,32,33} During early delithiation, the (003) peaks of both NC and NCZ move to a lower degree, which is due to the increased electrostatic repulsion between the TMO_2 slabs triggered by the removal of Li^+ .³⁴ Upon delithiation beyond ~ 52 mol %, the (003) peaks shift abruptly to a higher degree, which is mainly due to two reasons: (i) with more Ni^{3+} being oxidized to Ni^{4+} , the Ni–O bonds become more covalent, which in turn reduces the electrostatic repulsion between the O^{2-} ion in the adjacent TMO_2 slabs; and (ii) with the leaving of the pillaring Li^+ ions from the interlayers in both samples at a highly charged state, the lattice tends to collapse to maintain a smaller inter-slab distance.^{35–37} In spite of these similarities, NC and NCZ exhibit remarkable differences in their structural evolution at the deeply charged state. Specifically, an obvious phase separation of H2 and H3 is recorded in the NC sample upon charging to ~ 60 mol % delithiation, and such a coexistence of both phases lasts until the end of charging (80 mol % delithiation). Furthermore, as indicated by the white arrow, the intensity of the (003) peak sharply decreases upon the split of H2 and H3 phases, suggesting a sudden and structurally unfavorable transition.³⁸ In contrast, NCZ displays a pseudo-single-phase transition from H2 to H3 with a nearly indiscernible phase separation. The transition from H2 to H3

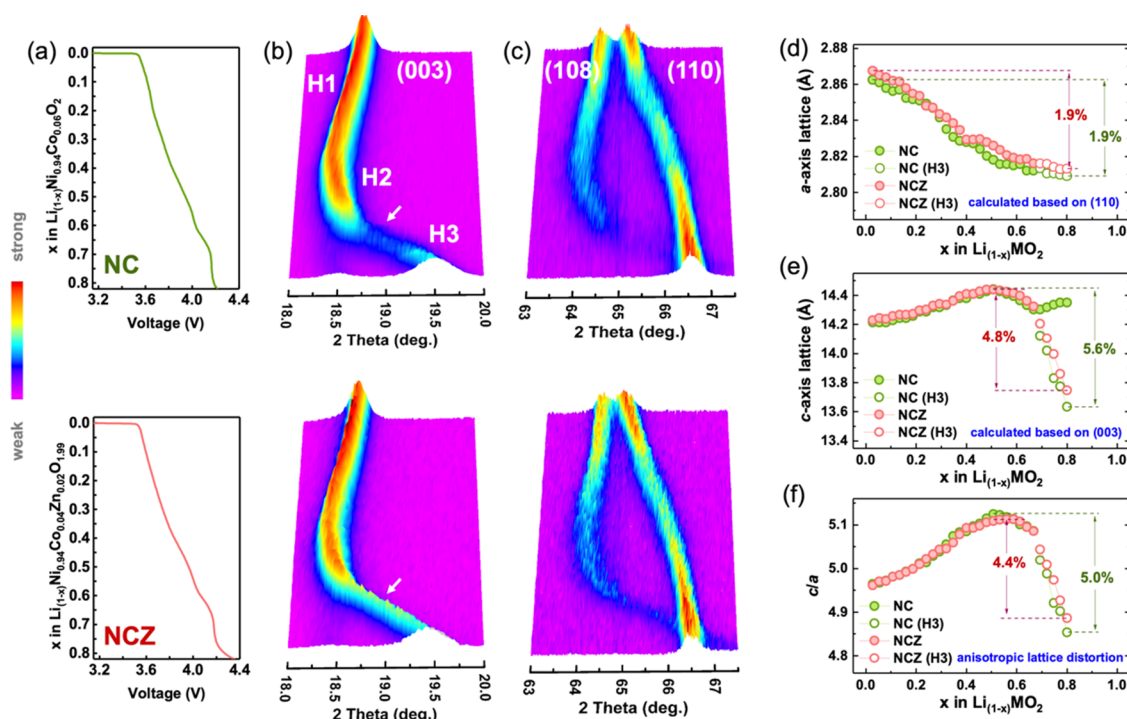


Figure 3. (a) Voltage profiles of NC and NCZ during delithiation up to the extraction of 80% Li^+ in half cells. Three-dimensional color plot of the diffraction patterns of (b) (003) peak and (c) (108)/(110) peaks of NC (top) and NCZ (bottom) during delithiation (80% cutoff). (d) a axis lattice parameter and (e) c axis lattice parameter evolution in NC and NCZ during delithiation and (f) calculated c/a value of both samples as a reference for the anisotropic lattice distortion.

(indicated by the white arrow) in NCZ is strikingly smooth without any “gap”, indicating a minor lattice mismatch between the H2 and H3 phases. Lattice parameter changes of both samples upon charging are exhibited in Figure 3d,e. As shown, the a lattice parameter of both samples shows an identical shrinkage of 1.9% during delithiation due to the oxidation of Ni^{3+} to Ni^{4+} during charging.²⁸ However, the lattice shrinkage of NC along the c axis (5.6%) is much larger than that of NCZ (4.8%). In addition, Figure 3f shows the c/a ratio of both samples upon delithiation, which is an indicator of anisotropic lattice distortion.³⁰ The degree of anisotropic lattice distortion in NCZ is 4.4%, which is 12% smaller than that of NC (5.0%), indicating a smaller lattice volume expansion/shrinkage in NCZ than in NC during lithiation/delithiation.

Figure 4a compares the SEM figures of NC and NCZ cathode electrodes after 500 cycles. Compared to the pristine sample (Figure S7), NC suffers from serious mechanical failure after cycling, with the majority of the secondary particles either shattered into small particles or fissured. On the other hand, although broken particles can also be found within the NCZ sample, most particles maintain the original spherical shape quite well, which is considered to result from the lower anisotropic lattice variation in NCZ as observed in the in situ XRD test. In addition, a low $I(003)/I(104)$ value (0.97) is found in the XRD pattern of the cycled NC electrode (Figure 4b), which indicates that a more layered rhombohedral phase to spinel/rock-salt phase transformation has occurred in NC during cycling.^{39,40} Such a serious structural degradation has been recognized to result from the propagation of electrode/electrolyte reactions from the surface of the particles to the bulk. With the Zn^{2+} insertion to mitigate particle cracking, NCZ could avoid serious electrolyte injection inside the

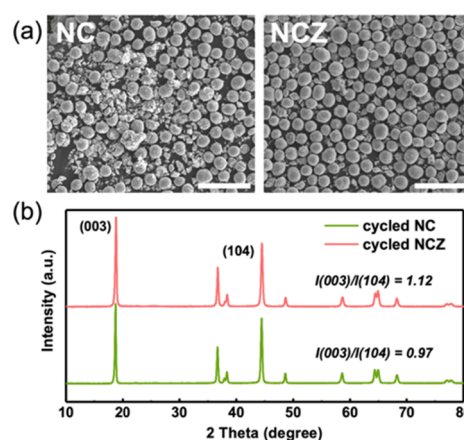


Figure 4. (a) SEM images and (b) XRD patterns of NC and NCZ samples harvested from pouch full cells cycled for 500 cycles. The scale bar is 50 μm .

secondary particles and hence thereby own a more preserved layered oxide structure with a high $I(003)/I(104)$ value (1.12). In summary, ex situ and in situ XRD and SEM examination results suggest that the incorporation of Zn^{2+} suppresses lattice distortions, promotes a smooth phase transition in high-Ni cathodes, and maintains the particle and structural integrity of high-Ni cathodes.

3.4. Electrode/Electrolyte Interphase Analysis. Apart from mechanical failure originating in the bulk, the high surface reactivity of ultrahigh-Ni cathodes at deep charge states also creates barriers for their practical applications. The surface instability can be evaluated by self-discharge experiments and monitoring the open circuit voltage (OCV) decay, during which the following reaction will occur:⁴¹

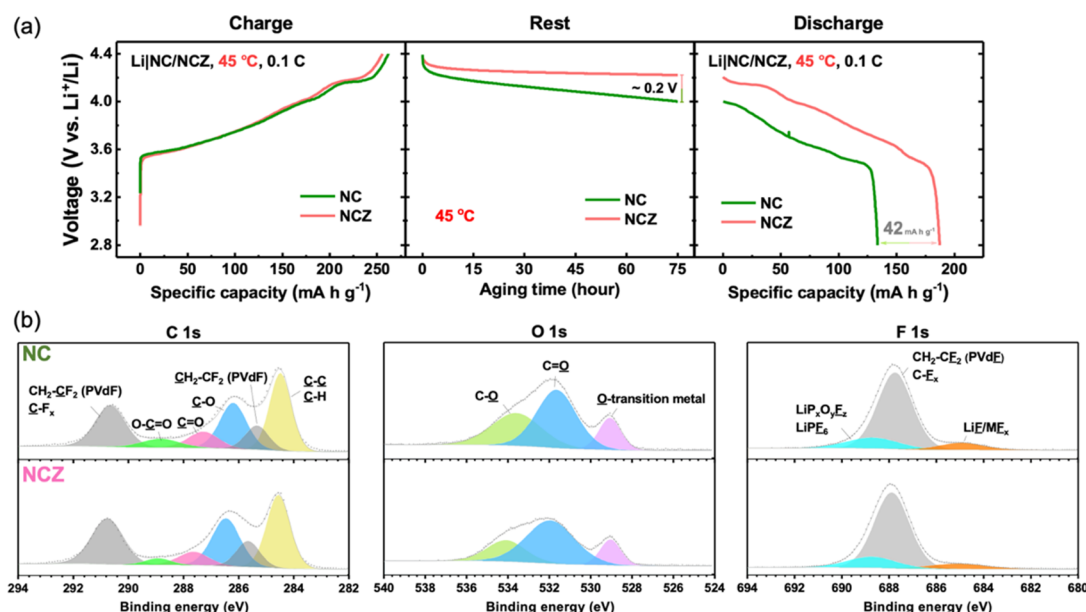
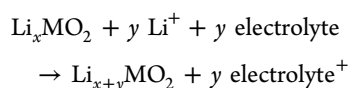


Figure 5. (a) Self-discharge profiles of NC and NCZ at 45 °C. (b) XPS patterns of both cathode electrodes harvested from pouch cells after 500 cycles: C 1s spectra, O 1s spectra, and F 1s spectra.



(M = Ni, Co, Zn, etc.)

Coin half cells with NC and NCZ were charged to 4.4 V at a C/10 rate and then rested for 75 h at 45 °C. Afterward, the cells were discharged to 2.8 V. As shown in Figure 5a, both NC and NCZ deliver an approximate charge capacity of >250 mA h g⁻¹. During the rest period at a high temperature, however, the OCV of the charged NC decays to 4.0 V, which is inferior to that of NCZ (4.2 V). Upon discharge, NCZ is able to deliver a capacity of 187 mA h g⁻¹, much higher than 144 mA h g⁻¹ delivered by NC. The self-discharge test results suggest that the introduction of Zn²⁺ can alleviate the surface reactivity of the high-Ni cathode and likely benefits the cathode–electrolyte interphase (CEI), in great agreement with the lower polarization observed in NCZ half cells.

Electrode/electrolyte interphase stability plays a critical role in determining the cycling performance of cathodes. To reveal the differences in the interphases of NC and NCZ, both cathodes were examined by XPS after 500 cycles in pouch full cells. Figure 5b displays the C 1s, O 1s, and F 1s XPS spectra of both cycled cathodes. As the C 1s and O 1s spectra show, the intensities of C–O and C=O peaks in the spectra of NCZ are weaker than those of NC, which indicates that less carbonate-based electrolyte decomposition occurs in NCZ.⁴² In the O 1s spectra, a higher intensity ratio of O–TM (representing the bulk layered cathode oxides) to C–O (or C=O) (representing the electrolyte decomposition products) is observed in NCZ, indicating a thinner CEI formed on NCZ during cycling.^{43,44} For the F 1s spectra, the intensity of the LiF/MF_x peak in NCZ is less than that in NC, which suggests that NCZ exhibits lower levels of transition-metal dissolution and electrode/electrolyte parasitic reactions compared to NC upon long-term cycling.⁴⁵ The XPS examinations on NC and NCZ cathodes illustrate the effectiveness of Zn²⁺ doping toward enhancing the interphase stability of the high-Ni

cathode and mitigating unfavorable interactions between the cathode and electrolyte.

It is worth noting that severe reactions between cathodes and electrolytes upon cycling not only cause the degradation of cathodes but also result in anode degradation.⁴⁶ Therefore, the graphite anodes paired with both cathodes after cycling are also examined by XPS. Zn 2p and Ni 2p XPS spectra obtained from the graphite anodes paired with NC and NCZ are presented in Figure S8. Compared to the NC graphite, the Ni 2p spectra detected from the graphite paired with NCZ exhibits a decreased intensity, implying that less Ni migration and deposition on the graphite surface occurs. As the Zn 2p spectra in Figure S7 show, there is no Zn detected in the graphite paired with NCZ after long-term cycling, indicating that the incorporation of Zn²⁺ into high-Ni cathodes has no adverse impact on the anode. Additionally, C 1s, O 1s, and F 1s, and P 2p XPS spectra of the harvested graphite anodes paired with NC and NCZ are shown in Figure S9. As seen, the intensity of the C–O and C=O peaks (representing the anode/electrolyte interaction products) on the graphite paired with NCZ is lower than that on the graphite with NC, while the C–C/C–H peaks (representing mainly bulk graphite) in the graphite paired with NCZ are slightly higher, indicative of a thinner SEI formed on the graphite paired with NCZ. Moreover, the electrolyte decomposition products, such as MF_x and Li_xPO_yF_z, are also found to be less concentrated on the graphite electrode paired with NCZ.

The impedance buildup of both samples after 10 and 500 cycles in pouch cells was evaluated by Nyquist impedance plots shown in Figure 6. Through impedance spectroscopy, complicated resistances or impedances from the cell components can be separated by applying a small voltage perturbation and recording the frequency responses.⁴⁷ Specifically, the first semicircle at the high-to-medium frequency is mainly indexed to the surface film resistance (R_{sf}) consisting of resistances from electrode–electrolyte interphases (EIs) at the anode and cathode surfaces. The second semicircle at the medium-to-low frequency mostly results from the charge-transfer resistance (R_{ct}) across the

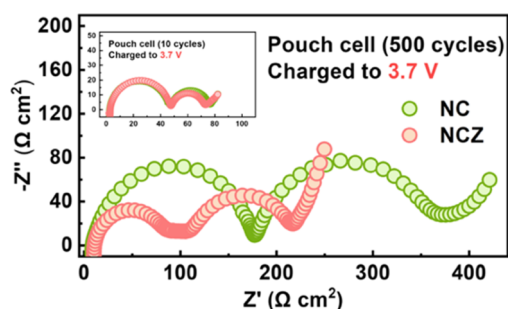


Figure 6. Nyquist impedance plots of NC and NCZ pouch full cells after 10 (inset) and 500 cycles. Both cells were controlled at 3.7 V.

cathode surface.^{48,49} Furthermore, the diameter of each semicircle represents the magnitude of the corresponding resistance. As shown in Figure 6, although NC and NCZ full cells initially deliver similar R_{sf} and R_{ct} values, NC suffers from a more rapid resistances increase (larger diameters of both semicircles) than NCZ after 500 cycles, which is in line with the electrochemical performance, mechanical failure, and XPS results discussed before. A smaller surface film resistance in the NCZ cell is believed to be the outcome of thinner but robust EEIs due to the mitigated side reactions and transition-metal dissolution as demonstrated by the XPS comparison. With respect to the charge-transfer resistance, NCZ shows a greatly reduced resistance as well, which is thought to be due to suppressed particle cracking and consequently mitigated contact loss and electrode/electrolyte side reactions as indicated by the SEM and XPS results (Figure 4a and Figure 5b). It is well recognized that the passivation of cathode particles with a layer of a high-resistance rock-salt phase would occur during cell operation as severe unwanted electrolyte–cathode reactions take place.⁵⁰ Beyond that, serious particle cracking continuously exposes more fresh cathode particles to the electrolyte, accelerating impedance buildup. NCZ, benefiting from less particle cracking and detrimental reactions, thus outperforms NC in terms of impedance growth.

To sum up, deleterious reactions between a high-Ni layered oxide cathode $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ and electrolyte are suppressed by the incorporation of 2 mol % of Zn^{2+} as indicated by the self-discharge experiment, XPS spectra, and the EIS plot. The combined results are in great agreement with the electrochemical performances and in situ XRD results, further validating the benefit of Zn-doping in stabilizing ultrahigh-Ni layered oxides.

3.5. Thermal Stability. Beyond energy density and cyclability, battery safety is drawing increasing attention. The poor thermal stability of Ni-based cathode materials (high-Ni cathode materials in particular) still remains a crucial hurdle to improving battery safety.⁵¹ DSC tests were employed to examine and compare the thermal abuse tolerances of both NC and NCZ. Both samples were tested under identical states of charge to ensure a proper comparison. As depicted in Figure 7, the exothermic peak temperature of NC is 180.7 °C accompanied by a heat release of 1380 J g⁻¹. In contrast, the exothermic peak temperature of NCZ is elevated to 187.1 °C with a significantly reduced heat release (795 J g⁻¹). These results indicate that the incorporation of Zn^{2+} is beneficial to increasing the thermal stability of high-Ni cathodes. Previous studies show that the highly charged high-Ni cathodes would experience the phase transitions from a layered structure to a spinel structure and finally to a rock-salt structure upon

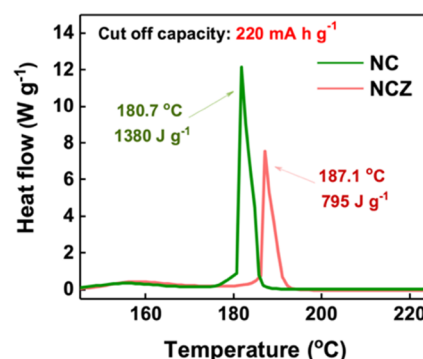


Figure 7. DSC profiles of NC and NCZ cathodes charged to 220 mA h g⁻¹ in half cells.

heating, which is accompanied by the release of oxygen.^{52–55} Due to the similarity of their ionic radius, some Zn^{2+} ions are proposed to enter the Li layer and play a role similar to Mg^{2+} in enhancing the structural stability of NCZ.²⁸ As a result, the delithiated NCZ requires a higher temperature to initiate the phase transition compared to NC. In addition, the decomposition of charged NCZ may result in the formation of ZnO , which reduces the amount of the released oxygen and thus suppresses the heat release.

4. CONCLUSIONS

In this work, we synthesized $\text{LiNi}_{0.94}\text{Co}_{0.04}\text{Zn}_{0.02}\text{O}_{1.99}$, a Zn-doped, ultrahigh-Ni cathode through hydroxide co-precipitation followed by calcination and comprehensively evaluated the effect of Zn. Compared to the pristine cathode $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$, the incorporation of 2% Zn element improves the cycling performance of NCZ without compromising the high capacity, regardless of pairing with the lithium-metal anode in half cells or the graphite anode in pouch full cells. The as-prepared NCZ delivers a high capacity of 218 mA h g⁻¹ at C/3 with a high, yet stabilized, average operation voltage at 3.82 V. NCZ also shows a capacity retention of 74% in pouch full cells, outperforming NC (62%). In situ XRD and post-mortem morphology analysis illustrates that Zn doping promotes a smooth phase transition, suppresses anisotropic lattice distortions, and maintains the mechanical integrity of the cathode secondary particles upon long-term cycling. Moreover, XPS and self-discharge results suggest that the presence of Zn in NCZ benefits the interphase stability of the cathode. DSC results indicate that NCZ exhibits thermal stability superior to NC. This work demonstrates the advantage of Zn doping in improving the performance of ultrahigh-Ni cathodes and provides a facile doping candidate for designing and synthesizing high-performance cathodes.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.1c01824>.

SEM images, XRD patterns and Rietveld refinement results, electrochemical data, XPS results, inductively coupled plasma atomic emission spectroscopy results, and crystallographic parameters (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Arumugam Manthiram – Materials Science and Engineering Program & Texas Materials Institute, The University of Texas at Austin, Austin, Texas 78712, United States;
orcid.org/0000-0003-0237-9563; Email: manth@
austin.utexas.edu

Authors

Zehao Cui – Materials Science and Engineering Program & Texas Materials Institute, The University of Texas at Austin, Austin, Texas 78712, United States

Qiang Xie – Materials Science and Engineering Program & Texas Materials Institute, The University of Texas at Austin, Austin, Texas 78712, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsami.1c01824>

Notes

The authors declare the following competing financial interest(s): The corresponding author (A. M.) is a co-founder of TexPower, Inc., a startup company focusing on cobalt-free cathode materials for lithium-based batteries.

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